

EC571: Digital VLSI Circuit Design**Homework 1 – Fall 2026****Total: 100 points****Due: September 23 at 11:59 pm ET****General Instructions**

Use this worksheet to complete the homework. Please print this worksheet, write your solutions on the worksheet, take a picture of the worksheet, and submit it through Gradescope. If you prefer, you can download the Homework 1 worksheet on your tablet, complete the worksheet on your tablet, and submit the completed worksheet through Gradescope.

Show all intermediate reasoning. A numerical answer, Boolean expression, or circuit diagram without explanation will receive limited credit.

For transistor-level questions, clearly indicate which transistors are ON/OFF for the input cases you analyze. For design questions, state your assumptions and justify your design choices.

You may use AI tools only for concept review. You may not use AI tools to produce final answers, final schematics, or final derivations. At the end of your submission, include one sentence disclosing whether and how you used AI tools.

C_{sw} = total switched capacitance

$$= N \times \cancel{\alpha} \times C$$

$\uparrow$ # of gates $\uparrow$ Activity factor $\uparrow$ Avg. C of single gate.
 ignore α - Yang Wu

NOTES PAGE
1 for Q1a

$$P_{dyn} = \alpha (N \cancel{\alpha} C) \times V_{dd}^2 \times f.$$

← ?? asked Piazza's Yang Wu

$$\frac{P'_{dyn}}{P_{dyn}} = \frac{\alpha \times (1.3 N \times 0.65 C) \times (0.8 V_{dd})^2 \times 1.25 f}{\alpha \times (N \times C) \times V_{dd}^2 \times f}$$

$$= \frac{\alpha 1.3 NC \times 0.64 V_{dd}^2 \times 1.25 f}{\alpha NC \times V_{dd}^2 \times f}$$

$$= 1.3 \times 0.64 \times 1.25$$

$$\frac{P'_{dyn}}{P_{dyn}} = 1.04.$$

- Increased transistor count increases P_{dyn} proportionally as P_{dyn} increases by total switched capacitance increasing increases by number of transistors in a block.
- Similarly, increasing average capacitor by transistor increases P_{dyn} as C is proportional to C_{sw} is proportional to P_{dyn} .
- P_{dyn} is proportional to V_{dd}^2 , i.e. it is 'proportional to the square'. This is due to the V_{dd} in the direct power approx. equation.
- frequency is directly proportional - also in P_{dyn} formula as simple multiplier.

Question 1: From Technology Scaling to Design Constraints [20 points]

This question asks you to reason about why VLSI design is not simply “make transistors smaller.”

A chip design team is comparing two technology generations. In the older generation, a digital block contains N transistors, has average switched capacitance C , operates at supply voltage V_{DD} , and runs at frequency f . In the newer generation, the team expects the following first-order changes:

$$N' = 2N \quad \# \text{ of TRANSISTORS}$$

$$C'_{\text{per transistor}} = 0.65 C_{\text{per transistor}} \quad \text{AVG SWITCHED CAPACITANCE}$$

$$V'_{DD} = 0.8 V_{DD} \quad \text{SUPPLY VOLTAGE}$$

$$f' = 1.25 f \quad \text{FREQUENCY}$$

Assume the activity factor is unchanged. Use the dynamic power approximation:

$$P_{\text{dyn}} = \alpha C_{\text{sw}} V_{DD}^2 f$$

where C_{sw} is the total switched capacitance.

(a) **Dynamic power scaling.** Derive the ratio

$$\frac{P'_{\text{dyn}}}{P_{\text{dyn}}}$$

Clearly show how transistor count, capacitance per transistor, voltage, and frequency each affect the result. [5 points]

See Notes Page 1 for
answers to Q1a :]

Energy per operation = $\frac{\text{dynamic power}}{\text{frequency}}$ ← ops per second

$$E = \frac{\propto NC(V_{dd}^2)f}{f}$$

f cancels out?
~~It might~~ could be misunderstanding the question - Ask Yang Wu.

$$E = \propto NC V_{dd}^2.$$

$$\frac{E'}{E} = \frac{\propto N' C' V'^2_{dd}}{\propto N C V^2_{dd}} = \frac{2N \times 0.65 C \times (0.8)^2 V_{dd}}{N C V^2_{dd}}$$

$$= 2 \times 0.65 \times 0.8^2 = 0.832$$

$$\frac{E'}{E} = 0.832.$$

Q1B,
Note page 7

(1c)

Answer from Pt. 1A: $\frac{P'_{dyn}}{P_{dyn}} = 1.04.$

$$\begin{array}{ccc} A' & = & 0.7A \\ \uparrow_{\text{new } A} & & \uparrow_{\text{old } A} \end{array}$$

$$\frac{P'_{dyn}/(0.7A)}{P_{dyn}/A} = \frac{1.04/0.7A}{1/A} = \frac{1.04}{0.7A} \times A = \frac{1.04}{0.7}$$

$$\frac{P'_{dyn}/A'}{P_{dyn}/A} = \frac{1.04}{0.7} = 1.4857 \text{ (4dp)}$$

$\frac{104}{70}$ $\uparrow$ 52/35

~~Notes~~ Notes Page 6
Q1c

- (b) **Energy per operation.** Assume the number of useful operations per second scales proportional to frequency only. Derive the ratio of energy per operation in the newer generation relative to the older generation. [4 points]

$$\frac{E'}{E} = 0.832$$

Working on note page 7

- (c) **Power density.** Suppose the chip area scales by $0.7\times$, meaning

$$A' = 0.7A.$$

Using your result from part (a), derive the ratio of power density:

$$\frac{P'_{\text{dyn}}/A'}{P_{\text{dyn}}/A}$$

See Notes page 6
for working -

[4 points]

$$\frac{P'_{\text{dyn}}/A'}{P_{\text{dyn}}/A} = \frac{1.04/0.7A}{1/A} \Rightarrow \frac{1.04}{0.7} = \frac{52}{35}$$

- ① The dynamic power approximation is higher - this on its own isn't good (unless the tradeoffs are worth it)
- ② The energy per operation has decreased from the old generation to the new - great! more efficient! however, why is the amount of energy used higher?
- ③ The power density has increased significantly, due to the (impressive) decrease in area. This means that the amount of energy (therefore heat) is far higher than the older version, requiring lots of cooling. This is not great.
- ④ Presumably the design is far more complex, to fit twice the number of transistors into 0.7x the area - but the newer technology is certainly not automatically better. As Prof. Joshi says, "it depends!"

Q1d, notes Pg 8

- ④ complex design - entire die image transferred @ once & resolution improving all the time.
& processed

Non Recurring cost $\uparrow$ every technology generation

Constraints:
 Resolution, alignment,
 Density. Parasitics.

Abstraction Device $\rightarrow$ circuit $\rightarrow$ gate $\rightarrow$ module $\rightarrow$ system

reasonable & acceptable amounts of failure $\rightarrow$ space to optimise

$\hookrightarrow$ Reduce frequency for ~~increased~~ reduced power consumption?

different bonding techniques - heat dissipation, cost

$\hookrightarrow$ Tape - automated bonding $\leftarrow$

$\hookrightarrow$ flip chip bonding

surface mount - higher density

Q1E notes
Pg 8

- (d) **Design interpretation.** Based on your answers, explain whether the newer technology is automatically "better." Your answer must discuss at least three of the following: performance, energy, power density, cooling, reliability, and design complexity. [4 points]

Previous answers:

$$\frac{P'_{dyn}}{P_{dyn}} = 1.04$$

$$\frac{E'}{E} = 0.832$$

$$\frac{\text{Power density}}{\text{Power density}} = \frac{52}{35} \approx 1.4857.$$

Please read
Notes Pg 8,
Q1D

- (e) **Course connection.** Identify two topics from the first EC571 lectures that help address the design challenges revealed by this scaling exercise. For each topic, explain in one sentence why it matters. [3 points]

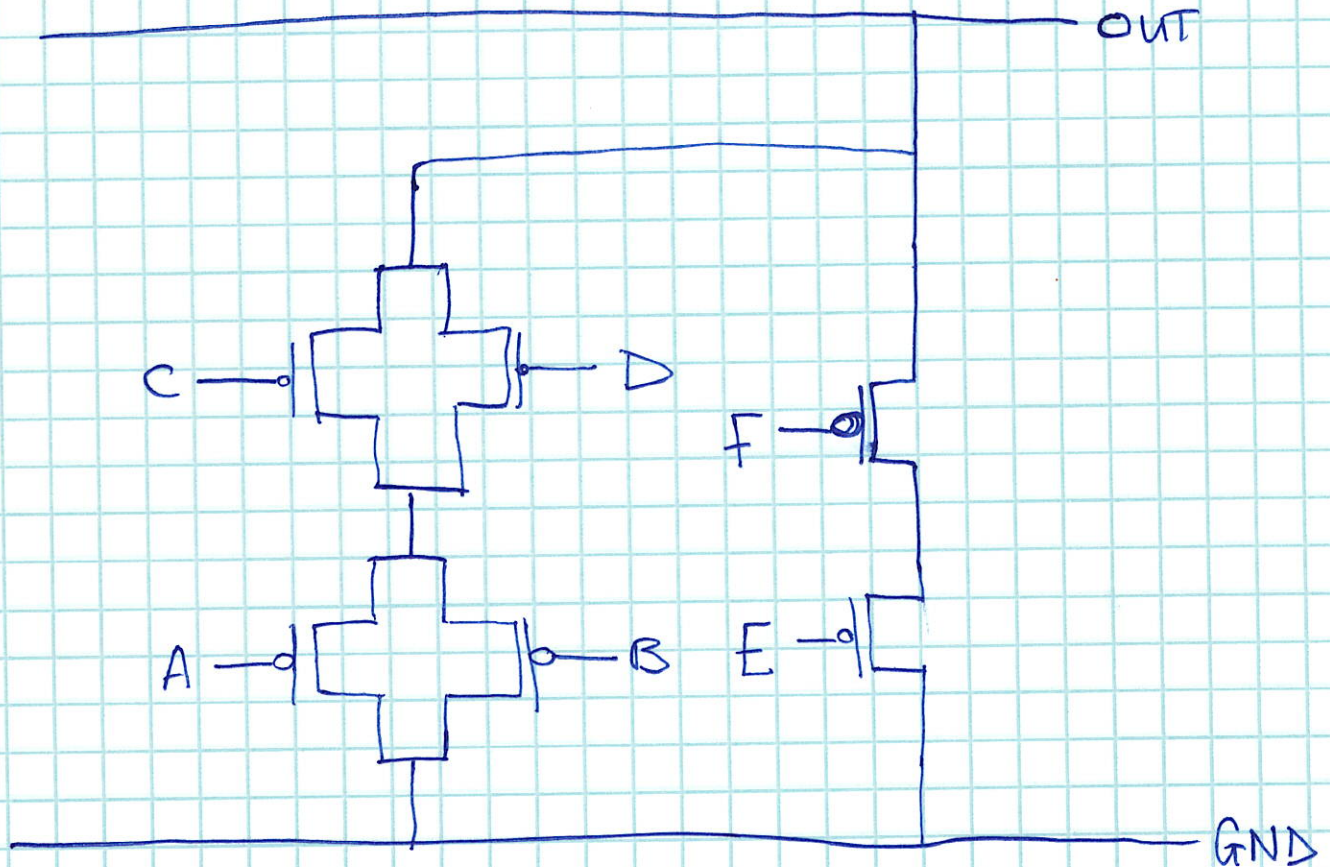
* Notes Pg 8, Q1E
For Notes taken whilst
Rewatching lectures

Reasonable & acceptable levels of failure allow for optimization at different levels of abstraction in design. Area ^{requirements} are unlikely to be relaxed over much but perhaps frequency could be reduced to reduce P_{dyn} and therefore Power density.

different bonding techniques, in the actual manufacture of the chip, could help mitigate the increased costs from increased design complexity.

$$F = (A \text{ OR } B) \text{ AND } (C \text{ OR } D) \text{ OR } E \text{ AND } F$$

$$\overline{F} = (A \text{ OR } B)(C \text{ OR } D) \text{ OR } E \text{ AND } F$$



Notes Page 2,
Q 2a

Question 2: Static CMOS Logic Implementation and Sizing [20 points]

Implement the following Boolean function as a single static complementary CMOS gate:

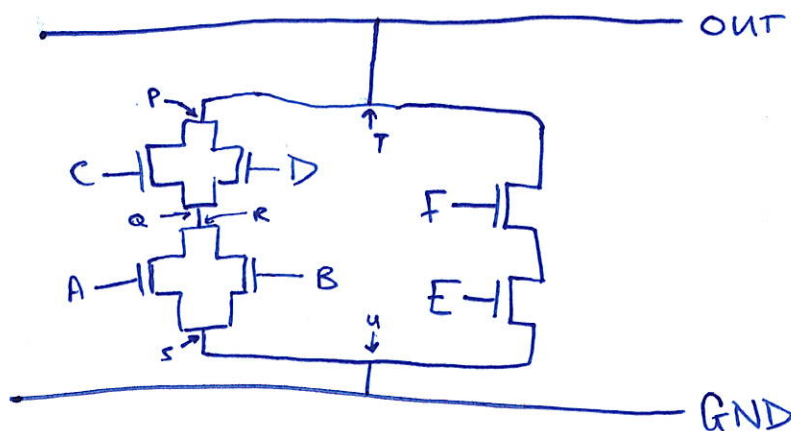
$$F = \overline{(A + B)(C + D) + EF}.$$

Do not decompose the function into multiple NAND/NOR/inverter gates. You may assume true and complemented input signals are not both available unless you explicitly add an inverter. For this problem, implement the function directly using only inputs A, B, C, D, E, F .

Assume:

- Unit width NMOS ON resistance is R_n .
- Unit width PMOS ON resistance is $2R_n$.
- Unit width NMOS input capacitance is C_g .
- Unit width PMOS input capacitance is $2C_g$.
- Transistor widths must be integer multiples of unit width.
- Ignore diffusion capacitance in the numerical parts.

- (a) **Pull-down network.** Draw the NMOS pull-down network that causes $F = 0$. Clearly label all transistor gates and internal nodes. [4 points]



$$x = \overline{F}$$

$$\text{When } F=0 \quad x=1$$

$$x = (A+B)(C+D) + EF$$

$$\overline{F} = (A+B)(C+D) + EF$$

FOR output to = 0 :

$$F=1 \text{ and } E=1$$

$$C \text{ OR } D = 1$$

$$A \text{ OR } B = 1$$

T: Branched connection to C OR D & A OR B, and to E and F.

P: Branched connection to C OR D

R: Branched connection to A OR B

Q: ~~AND~~ C OR D

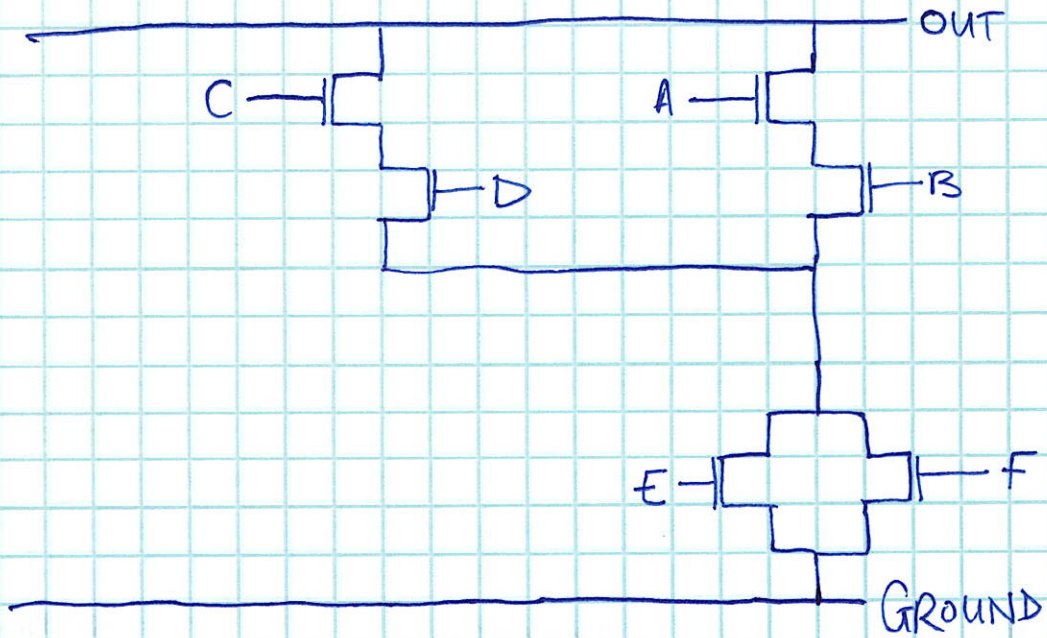
S: A OR B → GND

U: Branched connection to GND

More
Notes on
Note Page 2

NMOS STRONGER.
BOTTOM

if 0 then v.



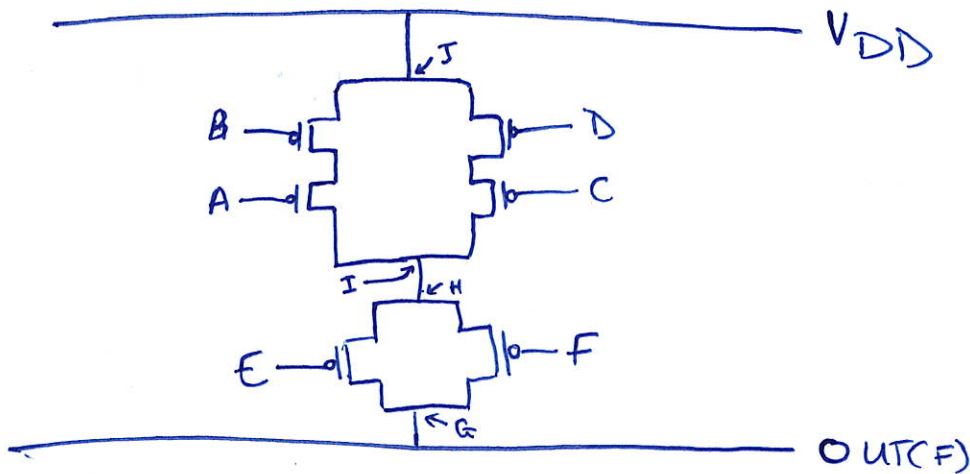
$A=0$
 $B=0$
~~AND~~
OR

$C=0$
 $D=0$

AND
 $E=0$
OR
 $F=0$

Notes Page 3
Q2b

- (b) **Pull-up network.** Draw the PMOS pull-up network as the dual of the pull-down network. Clearly label all transistor gates and internal nodes. [4 points]



PMOS STRONG IS TOP. $F=1$

Notes on Notes Page 3

- (c) **Correctness check using input vectors.** Choose one input vector for which $F = 0$. Identify the conducting path from V_{out} to ground. Choose one input vector for which $F = 1$. Identify the conducting path from V_{DD} to V_{out} . [4 points]

$F=0$ when $A=0$ $B=0$ $C=0$ $D=0$ $E=1$ $F=1$

Path: $V_{out} \rightarrow T \rightarrow F_{\text{NMOS TRANSISTOR}} \rightarrow E_{\text{NMOS TRANSISTOR}} \rightarrow U \rightarrow \text{GROUND}$

$F=1$ when $A=1$ $B=1$ $C=0$ $D=0$ $E=1$ $F=0$.

Path: $V_{dd} \rightarrow J \rightarrow B_{\text{PMOS TRANSISTOR}} \rightarrow A_{\text{PMOS TRANSISTOR}} \rightarrow I \rightarrow H \rightarrow E_{\text{PMOS TRANSISTOR}}$

$\rightarrow G \rightarrow V_{out}$.

- (d) **Worst-case resistance before sizing.** Assuming all transistors are minimum size, identify the worst-case pull-down path resistance and the worst-case pull-up path resistance in terms of R_n . [4 points]

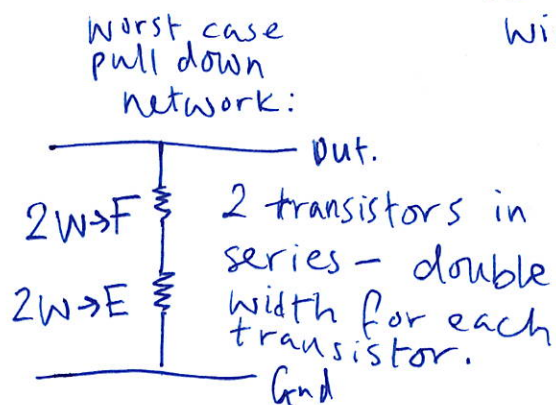
WORST CASE
PULL DOWN PATH:
 $V_{out} \rightarrow T \rightarrow F_{NMOS}$
 $\rightarrow E_{NMOS} \rightarrow U \rightarrow GND.$
2 NMOS TRANSISTORS W.
Resistance R_n :
WORST CASE PULL DOWN
PATH RESISTANCE = $2R_n$.

WORST CASE
PULL - UP PATH:
 $V_{out} \rightarrow G \rightarrow E_{PMOS} \rightarrow H \rightarrow I \rightarrow A_{PMOS} \rightarrow B_{PMOS} \rightarrow J$
 $E \rightarrow A \rightarrow B$: 3 PMOS in series.
 $3R_p$, $R_p = 2R_n$.
Worst case Pull up
Path resistance = $6R_n$.

- (e) **Transistor sizing.** Choose integer transistor widths so that the worst-case pull-down resistance is approximately R_n , and the worst-case pull-up resistance is approximately $2R_n$. Report the chosen width of every transistor.

Your answer must include a short explanation of how your sizing changes the input capacitance seen by each signal.

[4 points]

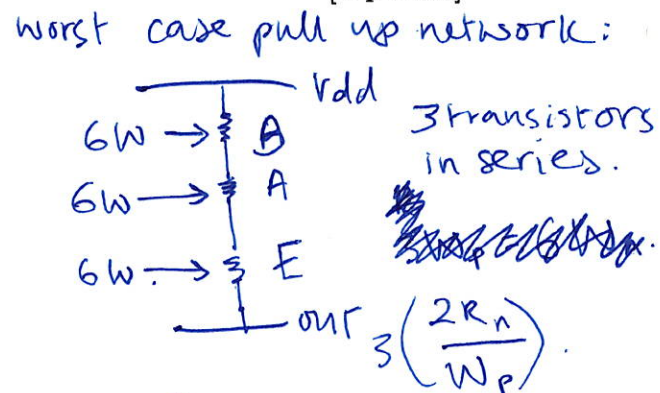


doubling
~~increasing~~ width increases
the input capacitance
(by 2x) because

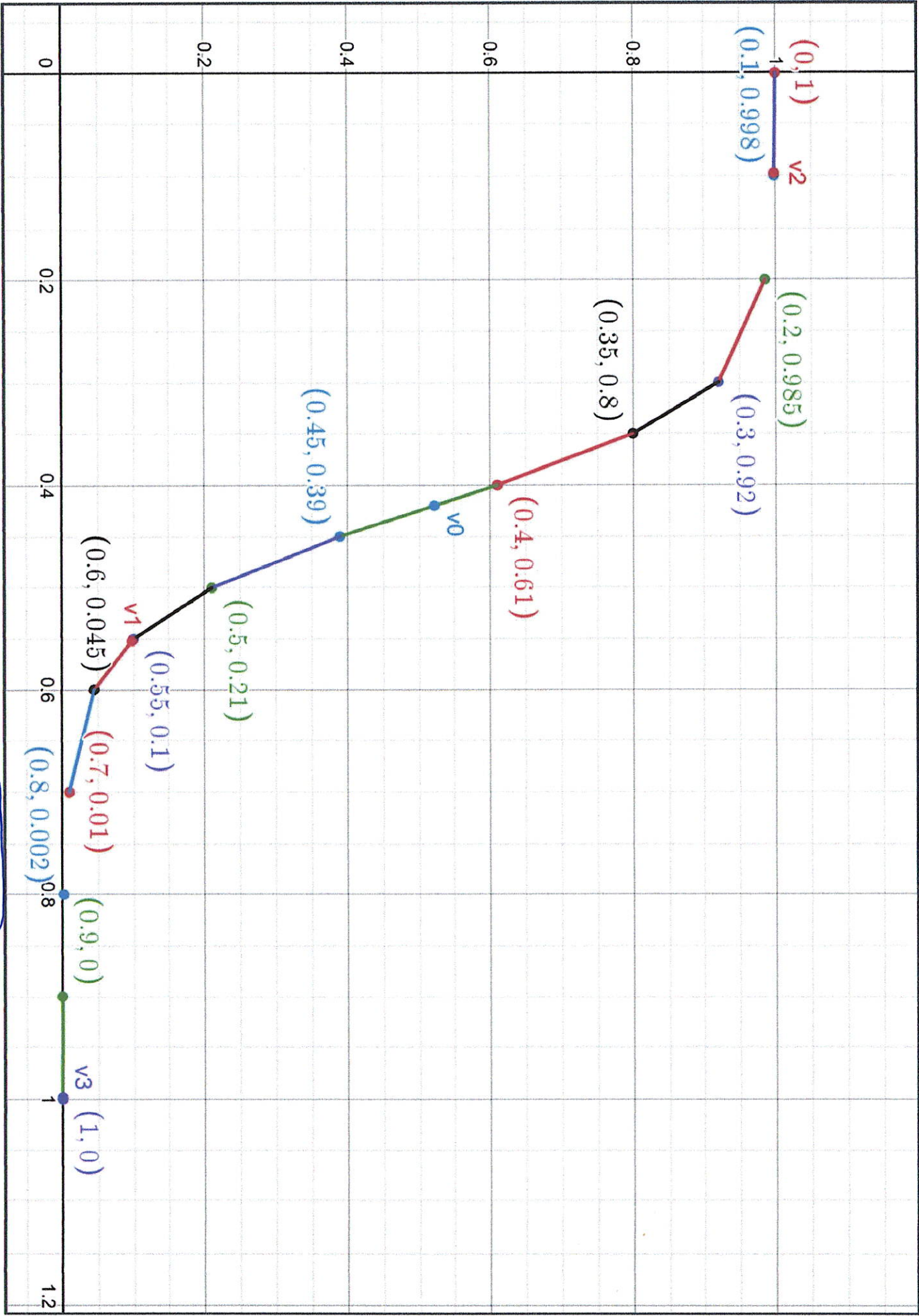
$$C_g \propto WL ?$$

↑
gate
capacitance

$W = \text{unit width}$



my increasing the
width of the ~~the~~ transistors
increases the input capacitance
seen by each
signal?



PRINTOUT 1

3 $(0.1, 0.998)$

4 $(0.2, 0.985)$

5 $(0.3, 0.92)$

6 $(0.35, 0.8)$

7 $(0.4, 0.61)$

8 $(0.45, 0.39)$

9 $(0.5, 0.21)$

10 $(0.55, 0.1)$

11 $(0.6, 0.045)$

12 $(0.7, 0.01)$

13 $(0.8, 0.002)$

14 $(0.9, 0)$

15 $(1, 0)$

16 $\approx$ approx curve

17 $y = -0.02x + 1 \{0 \leq x \leq 0.1\}$

18 $y = -0.65x + 1.115 \{0.2 \leq x \leq 0.3\}$

19 $y = -2.4x + 1.64 \{0.3 \leq x \leq 0.35\}$

20 $y = -3.8x + 2.13 \{0.35 \leq x \leq 0.4\}$

21 $y = -4.4x + 2.37 \{0.4 \leq x \leq 0.45\}$

22 $y = -3.6x + 2.01 \{0.45 \leq x \leq 0.5\}$

23 $y = -2.2x + 1.31 \{0.5 \leq x \leq 0.55\}$

24 $y = -1.1x + 0.705 \{0.55 \leq x \leq 0.6\}$

25 $y = -0.35x + 0.255 \{0.6 \leq x \leq 0.7\}$

26 $y = 0 \{0.9 \leq x \leq 1\}$

27 $\approx$ regenerating voltage inverters q

Log in or sign up to save your beautiful math!

PRINTOUT 2

V_{in} 's range of $0.40 \rightarrow 0.45$ is wholly contained by the corresponding V_{out} values of $0.610 \rightarrow 0.390$.

(V_{in}, V_{out}) $(0.4, 0.610)$ $(0.45, 0.390)$. (x, y) .

~~XXXXXXXXXX~~

$$m = \frac{0.39 - 0.61}{0.05} = -4.4.$$

$$y = mx + c.$$

$$c = 0.39 - (-4.4 \times 0.45) = 2.37.$$

$$V_m = -4.4(V_m) + 2.37. \text{ ie } V_{in} = V_{out}.$$

$$V_m + 4.4 V_m = 2.37$$

$$5.4 V_m = 2.37$$

$$V_m = \frac{2.37}{5.4} = 0.438 \bar{8} V$$

Notes Page 4
Q3A

Question 3: VTC, Noise Margins, and Regeneration [20 points]

A CMOS inverter has the following measured DC transfer characteristic. Assume:

$$V_{DD} = 1.0V.$$

V_{in} (V)	V_{out} (V)
0.00	1.000
0.10	0.998
0.20	0.985
0.30	0.920
0.35	0.800
0.40	0.610
0.45	0.390
0.50	0.210
0.55	0.100
0.60	0.045
0.70	0.010
0.80	0.002
0.90	0.000
1.00	0.000

FOR ENTIRE
QUESTION,
CHECKED
WORK AGAINST
GRAPH I MADE
ON DESMOS.
SHARED IN
PRINTOUTS 1&2.

Use only the data in the table. Do not assume the inverter is symmetric.

- (a) **Switching threshold.** Estimate the switching threshold V_M , defined as the point where

$$V_{in} = V_{out}.$$

Use linear interpolation between the two closest table entries.

[4 points]

See Notes Page 4
for working

$$V_M = 0.438 \text{ V}$$

$V_{in} \rightarrow V_{in}$

$$\frac{0.800 - 0.920}{0.05} = \frac{-0.120}{0.05} = -2.4 \quad \begin{array}{l} \nwarrow \text{slope} \\ 0.3 \rightarrow 0.35 \end{array}$$

$$\frac{0.610 - 0.8}{0.05} = -3.8 \quad \begin{array}{l} \nwarrow \text{slope} \\ 0.35 \rightarrow 0.4 \end{array}$$

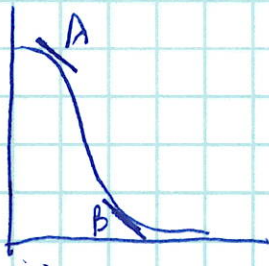
$$\frac{(0.390 - 0.610)}{0.05} = -4.4 \quad \begin{array}{l} \nwarrow \text{slope} \\ 0.4 \rightarrow 0.45 \end{array}$$

$$\frac{0.210 - 0.39}{0.05} = -3.6 \quad \begin{array}{l} \nwarrow \text{slope} \\ 0.45 \text{ to } 0.5 \end{array}$$

$$\frac{0.1 - 0.21}{0.05} = -2.2 \quad \begin{array}{l} \nwarrow \text{slope} \\ 0.5 \text{ to } 0.55 \end{array}$$

$$\frac{0.045 - 0.1}{0.05} = -1.1 \quad \begin{array}{l} \nwarrow \text{slope} \\ 0.55 \text{ to } 0.6 \end{array}$$

Notes Page 5
Q3B



Point A: Between V_{in} 0.2 and 0.35.
(V_{IL})

$M = -0.65 \rightarrow 1 \rightarrow M = -2.4$

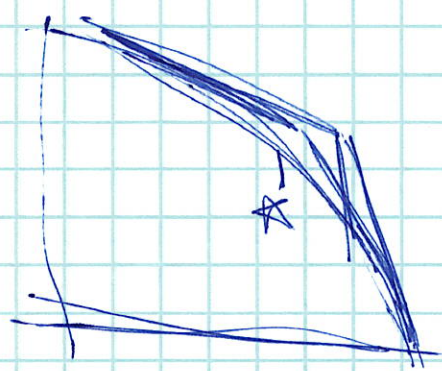
$M = -0.65$	$M = -0.65$	$M = -2.4$
0.2	0.25	0.3 0.35

$-2.4 + 0.65 \Rightarrow \Delta M = -1.75$

$-1 / -1.75 \approx 0.57$, which is less than $2/3$, so I can assume the point where

$m = -1$ for section A lies between $V_{in} = 0.2$ and $V_{in} = 0.3$. Furthermore,

~~0.2 + (1/1.75) * 0.15~~
= approx 0.28571



SIMILARLY, for POINT B: Between V_{in} 0.55 and 0.7
(V_{IH})

V_{in} 0.55 $\rightarrow$ V_{in} 0.7
 $-1.1 < 1 < -0.35$

$M = -1.1$	$M = -0.35$	$M = -0.35$
0.55	0.6 0.65	0.7

WRONG segments!

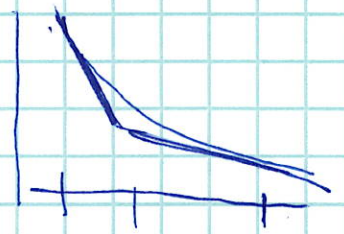
~~0.55 + (1/1.45) * 0.15~~

$M = -1.1$	$M = -0.35$	$M = -0.35$
0.55	0.6 0.65	0.7

$-0.35 \neq 1.1 \Rightarrow \Delta M = -1.45$

~~0.55 + (1/1.45) * 0.15~~

$V_{IH} = 0.55 + \frac{1}{1.45} \times 0.15 = 0.653448...$



Notes Page 9
Q3c

(b) **Local slope estimation.** Compute the approximate slope

$$\frac{\Delta V_{out}}{\Delta V_{in}}$$

for each interval between $V_{in} = 0.30V$ and $V_{in} = 0.60V$. Identify the interval with the largest magnitude slope. [4 points]

The interval with the largest magnitude slope is $V_{in} (0.4 \rightarrow 0.45)$ with a slope of -4.4

Please see notes page 5.

(for approx. slopes for each interval & intermediate calculations)

(c) **Noise-margin estimation.** Estimate V_{IL} and V_{IH} using the condition

$$\frac{dV_{out}}{dV_{in}} = -1.$$

Because the data is discrete, you must explain which intervals you use to estimate V_{IL} and V_{IH} . Then compute

$$NM_L = V_{IL} - V_{OL}$$

and

$$NM_H = V_{OH} - V_{IH}.$$

Use $V_{OH} = 1.0V$ and $V_{OL} = 0V$.

[5 points]

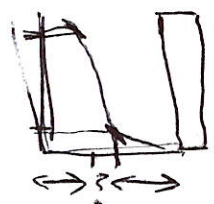
See Notes Page 9 for V_{IL} & V_{IH} calc.

$$V_{IL} = \text{approx } 0.28571$$

$$V_{IH} = \text{approx } 0.65345.$$

$$NM_L = V_{IL} - 0_{V_{OL}} = 0.28571V.$$

$$NM_H = V_{OH} - V_{IH} = 1 - V_{IH} = 0.34655V.$$



- (d) **Regeneration through a chain of inverters.** Suppose the input to the first inverter in a chain is

$$V_0 = 0.42V.$$

Use interpolation from the table to estimate

$$V_1, V_2, V_3,$$

where V_i is the output of the i -th inverter. Based on these values, explain whether the signal regenerates toward a legal logic value. [5 points]

$$V_{0\text{in}} = 0.42. \quad V_{0\text{-out}} = 0.552 \quad (y = -4.4x + 2.37)$$

$$V_{1\text{in}} = 0.552 \quad V_{1\text{-out}} = 0.0978 \quad (y = -1.1x + 0.705)$$

$$V_{2\text{in}} = 0.0978 \quad V_{2\text{-out}} = 0.998044 \quad (y = -0.02x + 1)$$

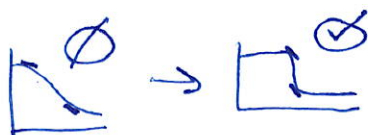
$$V_{3\text{in}} = 0.998044 \quad V_{3\text{-out}} = 0 \quad (y = 0.)$$

the signal oscillates closer and closer to 0 and 1 (magnitude of the) but not towards a legal logical value it depends on the number of inverters in the chain?

USING M values from Q3B, Pg 5 notes.

- (e) **Design interpretation.** If this inverter were used in a noisy environment, would you prefer a steeper or shallower VTC transition region? Explain your answer in terms of noise margins and regeneration. [2 points]

I would prefer a steeper transition region because I would want to reduce the range of voltages that would regenerate poorly & would want to maximise the noise margins by increasing the 'allowable' ranges of voltages.



Question 4: Inverter-Chain Delay Optimization with Non-Ideal Constraints [20 points]

A minimum-size inverter has input capacitance

$$C_{in,min} = 10fF.$$

You must drive a load

$$C_L = 6.4pF.$$

Use the normalized inverter delay model

$$d_i = f_i + \gamma$$

where

$$\gamma = 1.$$

For an inverter chain, the effective fanout of stage i is

$$f_i = \frac{C_{in,i+1}}{C_{in,i}}$$

for intermediate stages, and

$$f_N = \frac{C_L}{C_{in,N}}$$

for the final stage.

(a) **Electrical effort.** Compute the total electrical effort

$$F = \frac{C_L}{C_{in,min}}.$$

↑
fanout

[3 points]

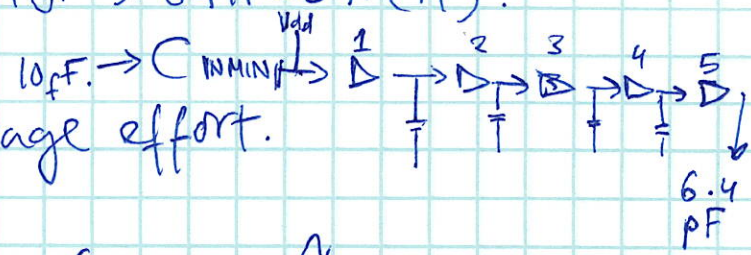
$$\frac{6.4pF}{10fF} = \frac{6400fF}{10fF} = \underline{640}$$

Q4c

$N=5$. equal stage effort is $3.6411284 \cdot (A)$.

$$f_N = f_5 = \frac{C_L}{C_{IN,5}}$$

$f = \text{stage effort}$.



$$A = \frac{6.4 \text{ pF}}{C_5} \rightarrow C_5 = \frac{6.4 \text{ pF}}{A} \approx 1.7576969 \text{ pF}$$

$$C_4 = \frac{C_{IN,5}}{A} \approx 0.4827341 \text{ pF}$$

$$C_3 = \frac{C_{IN,4}}{A} = 0.1325782 \text{ pF}$$

$$C_2 = \frac{C_{IN,3}}{A} = 0.0364113 \text{ pF}$$

$$C_{IN,1} = \frac{C_{IN,2}}{A} = 0.0100000 \text{ pF} = C_{IN,1}$$

(hopefully equals $C_{in,min}$)

$$C_{IN,MIN} = C_{IN,1} = 10 \text{ fF}$$

$1 \text{ pF} = 1000 \text{ fF}$



Notes Pg 11, Q4c

- (b) **Ideal number of stages.** Using the rule of thumb that a good stage effort is near 4, estimate the best number of inverter stages. Justify your answer. [4 points]

5 gates/stages -

stage effort per inverter closer to 4 when 5 gates than 4 gates

Working on Notes Page 10
Reasoning on Notes Page 10

- (c) **Continuous sizing.** For your chosen number of stages N , compute the equal stage effort

$$f = F^{1/N}.$$

Then compute the input capacitance of each inverter stage in the chain, starting with the minimum-size inverter. [5 points]

$$N = 5. \quad F^{1/5} = f = 3.6411284$$

$\nearrow 640^{1/5}$ (see pg 10, Q4B for working for this)

see notes page 11
For working please!
 $\uparrow$
\$ answers

4d)

Notes Pg 12

~~$$F_1 = \frac{C_{IN1}}{C_{IN MIN}} = \frac{10 \text{ fF}}{10 \text{ fF}} = 1$$~~

~~F_2~~ Start from end!

Effective fanout

$$F_5 = \frac{6400 \text{ fF}}{C_{IN5}} = \frac{6400}{1280} = 5 \text{ electrical fanout.}$$

$$F_4 = \frac{C_{IN5}}{C_{IN4}} = \frac{1280}{320} = 4 \text{ electrical fanout.}$$

$$F_3 = \frac{C_{IN4}}{C_{IN3}} = \frac{320}{80} = 4 \text{ electrical fanout.}$$

$$F_2 = \frac{C_{IN3}}{C_{IN2}} = \frac{80}{20} = 4 \text{ electrical fanout.}$$

$$F_1 = \frac{C_{IN2}}{C_{IN1}} = \frac{20}{10} = 2 \text{ electrical fanout.}$$

total normalized delay

$$d_i = f_i + \gamma, \gamma = 1.$$

$$\begin{aligned} d_1 &= 2+1=3 \\ d_2 &= 4+1=5 \\ d_3 &= 4+1=5 \\ d_4 &= 4+1=5 \\ d_5 &= 5+1=6 \end{aligned}$$

sums to: 24 tau

↑
unit of time?

$$15+9=24$$

ask yang Wu

- (d) **Discrete sizing constraint.** Now assume that only the following inverter input capacitances are available:

10fF , 20fF , 40fF , 80fF , 160fF , 320fF , 640fF , 1280fF , 2560fF .

Choose a practical inverter chain using only these values. Compute the effective fanout of each stage and the total normalized delay. [4 points]

$$C_1 = 0.01\text{pF} = 10\text{fF} \rightarrow \text{uses } 10\text{fF inverter.}$$

$$C_2 = 0.0364\text{pF} = 36.4\text{fF} \rightarrow \text{uses } \overset{20}{\cancel{10}}\text{fF inverter.}$$

$$C_3 = 0.1326\text{pF} = 132.6\text{fF} \rightarrow \text{uses } \overset{80}{\cancel{20}}\text{fF inverter.}$$

$$C_4 = 0.4827\text{pF} = 482.7\text{fF} \rightarrow \text{uses } 320\text{fF inverter.}$$

$$C_5 = 1.7577\text{pF} = 1757.7\text{fF} \rightarrow \text{uses } 1280\text{fF inverter}$$

Please see Pg 12 Notes for calculation of fanout for all stages

*"It depends" - Prof Joshi I assume "Excessive capacitance leads to higher inrush current, size & cost" flying-tech.com/blog/

- (e) **Energy-delay design tradeoff.** Suppose a second design uses one extra inverter stage compared with your answer in part (d), and produces almost the same delay but uses larger total gate capacitance. Explain why a designer might reject the second design even if its delay is similar. capacitor-power-supply-design.

Your answer must discuss both dynamic energy and input loading on the previous logic stage. [4 points]

A designer may reject design 2 due to added cost/area required by the extra inverter. Additionally, an extra gate would add another $(f_i + 1)$ tau to the total normalised delay, slowing the inverter chain. Also, the energy lost in fanout increases both by virtue of adding a gate and the larger total gate capacitances increasing the time needed to fill up each inverter & propagate the signal.

Question 5: Logical Effort and Design Tradeoffs [20 points]

You need to design a logic path that computes a control signal Y and drives a large downstream load. The Boolean function is:

$$Y = \overline{ABCD}.$$

The signal A arrives from a minimum-size inverter whose input capacitance is:

$$C_{in,0} = 10 \text{ fF}.$$

The final output Y must drive:

$$C_L = 1.28 \text{ pF}.$$

Assume the following logical effort model:

$$d_i = g_i f_i + p_i$$

where g_i is the logical effort of gate i , f_i is the electrical fanout of gate i , and p_i is the parasitic delay of gate i .

Use the following logical-effort and parasitic-delay values:

$$\text{Inverter: } g = 1, p = 1.$$

$$n\text{-input NAND: } g = \frac{n+2}{3}, p = n.$$

$$n\text{-input NOR: } g = \frac{2n+1}{3}, p = n.$$

You are considering two implementations.

Implementation 1: Use one 4-input NAND gate to compute

$$Y = \overline{ABCD},$$

followed by a chain of inverters if needed.

Implementation 2: First compute two intermediate NAND outputs:

$$X_1 = \overline{AB}, \quad X_2 = \overline{CD},$$

then combine them using a 2-input NOR gate:

$$Y = \overline{X_1 + X_2}.$$

You may add inverters after Y as output buffers if needed. You may also insert an inverter on an internal signal only if you explain why it is logically required.

5a)

Pg 13 of Notes

$$y = \overline{ABCD}$$

$$y = \overline{AB + CD} = \overline{A} + \overline{B} + \overline{C} + \overline{D}$$

if we assume that implementation 2's logic is an unchanging block*, then:

$$\left. \begin{array}{l} y_{2,old} = \overline{X_1 + X_2} \\ y_{2,old} = y_1 \end{array} \right\} y_{2,new} = \overline{y_{2,old}} = \overline{\overline{(X_1 + X_2)}}$$

use a NOT gate (inverter) to correct implem. 2.

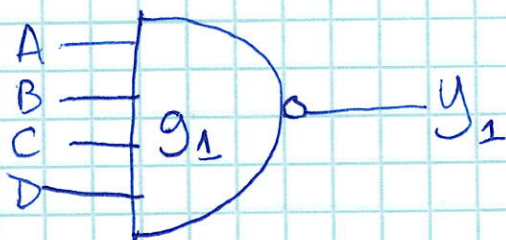
$$\begin{aligned} y_{2,new} &= \overline{\overline{X_1 + X_2}} = \overline{y_{2,old}} & X_1 &= \overline{AB} \\ &= X_1 + X_2 & X_2 &= \overline{CD} \\ &= \overline{AB} + \overline{CD} \\ &= \overline{ABCD} = y_1 \end{aligned}$$

*unless 'logically required'? NOT technically required, can complete without it?

5b

Implementation 1:

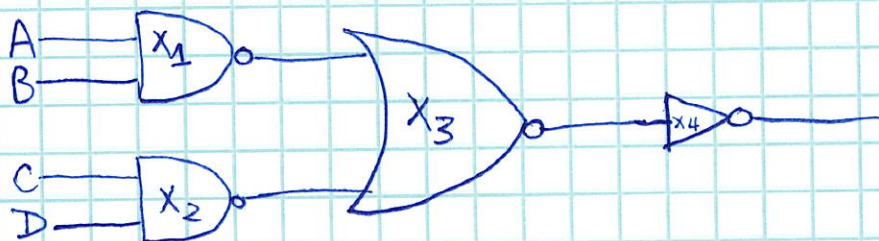
Notes Pg 14



$N_{\text{input NAND}} : g_1 = \frac{n+2}{3} = \frac{4+2}{3} = 2.$

$G = \prod_i g_i = 2.$

Implementation 2 (corrected)



X_1 is 2 input NAND. $n=2$. $\frac{2+2}{3} = \frac{4}{3}.$

X_2 is 2 input NAND. $n=2$. $\frac{2+2}{3} = \frac{4}{3}.$

X_3 is 2 input NOR. $n=2$, $\frac{2 \times 2 + 1}{3} = \frac{5}{3}.$

X_4 is inverter. $g=1$

$G = \frac{4}{3} \times \frac{4}{3} \times \frac{5}{3} \times 1 = \frac{80}{27}.$

- (a) **Logic correctness.** Verify whether Implementation 1 and Implementation 2 compute the same function. For Implementation 2, explicitly simplify

$$\overline{\overline{AB} + \overline{CD}}$$

and state whether it equals $\overline{ABCD}$. If it does not, propose the smallest correction to Implementation 2 using NAND/NOR/inverter gates. [4 points]

De Morgan!

$$y = \overline{X_1 + X_2} = \overline{\overline{AB} + \overline{CD}} \quad \begin{matrix} X_1 = \overline{AB} \\ X_2 = \overline{CD} \end{matrix}$$

$$\hookrightarrow y = (\overline{X_1}) \cdot (\overline{X_2})$$

$$y = (\overline{\overline{AB}}) \cdot (\overline{\overline{CD}})$$

$$y = (AB) \cdot (CD)$$

See Pg 13 of notes for working out smallest correction to implementation 2

$\hookrightarrow y = ABCD$ not same as $\hookrightarrow y = \overline{ABCD}$
Implementation 2 Implementation 1

- (b) **Path logical effort.** For the corrected versions of the two implementations, identify the path from input A to output Y. Compute the path logical effort

$$G = \prod_i g_i$$

for each implementation before adding output-buffer inverters.

[4 points]

Implementation 1: $y_1 = \overline{ABCD}$ Implementation 2 (New): $y_2 = \overline{\overline{X_1 + X_2}}$

$$X_1 = \overline{AB}$$

$$X_2 = \overline{CD}$$

Please see Notes Pg 14 for working.

$$\text{Imp. 1: } G = \frac{6}{3} = 2.$$

$$\text{Imp 2: } G = \frac{80}{27}$$

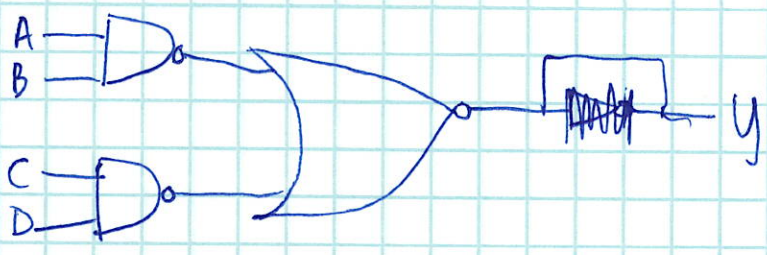
5c

Notes Pg 15

Implementation 1, no branching effort.

$$F = \frac{C_L}{C_{in0}} = \frac{1280 \text{ fF}}{10 \text{ fF}} = 128 \leftarrow \text{path electrical effort.}$$

Implementation 2 (corrected) ^{NOT}. No Branching outputs.



Path electrical effort: product of the effort per gate.

first gate 10 fF
last output 1280 fF

$$F = \frac{C_L}{C_{in0}} = \frac{1280 \text{ fF}}{10 \text{ fF}} = 128 ??$$

$$H = GFB$$

↑ ↑ ↑
gate path Branch

AWARE MISTAKE EXISTS - DON'T KNOW WHAT IS CORRECTION?

Imp. 1: $H = 2 \times 128 \times 1 = 256$

$$G = 2$$
$$F = 128$$
$$B = 1$$

$$N = \log_4(256) = 4.$$

Implementation 2:

$$G = \frac{80}{27} \quad F = 128$$
$$B = 1$$

$$H = \frac{80}{27} \times 128 = \frac{10240}{27} = 379.26$$

$$\log_4(H) \approx 4.284.$$

rounds to 4.

- (c) **Path effort and best number of stages.** For each implementation, compute the path electrical effort

$$F = \frac{C_L}{C_{in,0}}.$$

Assume no branching effort for Implementation 1. For Implementation 2, include any branching effort that appears on the path from A to Y . Then compute

$$H = GFB$$

and estimate the best number of stages using

$$N \approx \log_4 H.$$

Round N to a reasonable integer and justify your choice.

[4 points]

Please see Notes Pg 15 for
working & answer.

- (d) **Gate sizing.** For the implementation you believe is better, choose gate input capacitances so that each stage has approximately equal stage effort

$$h = H^{1/N}.$$

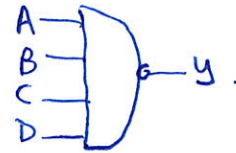
Work backward from the load and report the input capacitance of each stage on the path from A to Y. If you add output-buffer inverters, include them in the sizing. [5 points]

I believe that imp. 1 is better. ~~than the incorrect imp. 2.~~ Notes

Imp 1 N stages from 5c Page 15: $N = 4$. $H = 256$.

Individual stage effort $h = 256^{1/4} = 4$.

$$C_L = 1280 \text{ fF}$$



input gate capacitance = 10 fF.

4 stages within Imp 1 Nand gate? Am certain there is an error but. unclear where in my work & assumptions it is.

- (e) **Design-review recommendation.** Write a short design-review recommendation choosing between the two implementations. Your answer must discuss at least three of the following:

- logic correctness,
- number of stages,
- logical effort,
- parasitic delay,
- branching effort,
- input capacitance seen by signal A,
- robustness of using a large complex gate versus smaller gates.

↳ Not the corrected implementation, the original!

A purely qualitative answer such as "fewer stages is better" will receive limited credit. [3 points]

I would Recommend Imp. 1 as Imp. 2 is not logically correct (it isn't equivalent to Imp 1), contains fewer stages, less parasitic delay (just one gate!) has an increased robustness due to being a self-contained gate, and Requires less path-logical effort (2 versus Imp 2's $\frac{80}{27}$).

I have not used AI or LLMs, except for the occurrences where I have googled formulae and gotten to see ~~see~~ ^{scroll past} the LLM widget before the actual search results (wish it gave me an option to turn it off!).